Bob Ma

Senior Machine Learning Engineer · ML Pipelines & MLOps · Data Platforms · Production ML

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WORK RIGHTS NZ Permanent Resident AVAILABLE Immediately

§ 01 SUMMARY

Machine learning engineer in Auckland with 6+ years in production ML and 18+ years in software and data engineering. Holds a PhD from AUT (machine learning for privacy preservation) and an MSc in computer software and theory. Specialises in the *data pipelines and automation* that keep models running in production: ETL on AWS and Snowflake, Spark with Airflow, CI/CD with GitHub Actions and Concourse, MLflow tracking and Azure ML endpoints. Most recently, at Resideo, built VLM-based business assessment for small and medium-sized business (SMB) customers, from store data through to POS, payment and alerting systems. Experience spans smart-home, transport, retail and construction. Inventor on three patents and reviewer for AAAI, ACM TOMM and IEEE journals.

§ 02 CORE SKILLS / STRENGTHS

ML PIPELINES & MLOPS
Automated training, evaluation and deployment pipelines with CI/CD (GitHub Actions, Concourse), MLflow tracking and validation gates before release.

DATA PLATFORMS
ETL on AWS (SageMaker, Glue, S3) and Snowflake; Spark with Airflow; vector databases; SAP / ERP integration.

PRODUCTION ENGINEERING & DEVOPS
Python and C++ services behind REST APIs, containerised with Docker and Kubernetes and covered by automated unit and stress tests.

SMB BUSINESS FLOWS
Turning store data (photos, shelf images, camera video, POS and inventory) into assessments that feed POS, payment, alerting and ticketing systems.

VLMS, LLMS & AGENTS
VLM business assessments configured per customer, multimodal LLM inference behind REST APIs, and the Claude Agent SDK for code review and data processing.

FORECASTING & APPLIED ML
Demand, price and sales forecasting; driving-behaviour modelling (LSTM, k-means).

§ 03 WORK EXPERIENCE

Senior Machine Learning Engineer

Mar 2024 — Oct 2026

Resideo Inc. (Honeywell Home) · NYSE:REZI

- *Production ML pipelines and VLM solutions for SMB customers*
- Designed and optimised multimodal LLM inference (LLaMA3, GPT, CLIP) and connected model outputs to backend decision systems through REST APIs and edge–cloud pipelines on Azure.
- Owned the end-to-end model pipeline: PyTorch training, ONNX conversion, MLflow tracking, Docker builds released through GitHub Actions, and serving on Azure ML endpoints with a vector database.
- Used the Claude Agent SDK for automated code review in CI and for parts of the user-flow data processing, including building vector-database indexes.
- Built store-data pipelines for small and medium-sized business (SMB) customers, combining shopfront, shelf and in-store camera images with POS and inventory data.
- Used vision–language models (VLMs) for business assessments such as merchant verification, operating risk, shelf compliance and security-event summaries.
- Connected the results to POS, payment, alerting and ticketing systems, and met each customer's needs through configurable rules and prompt templates.
- Built automated validation and CI/CD testing for production reliability.

Stack: Python · PyTorch · ONNX · LLMs / VLMs · Claude Agent SDK · **ML DevOps:** GitHub Actions · Docker · MLflow · Azure ML · Vector database

Senior Machine Learning Engineer

Mar 2022 — Mar 2024

EROAD Ltd · ASX:ERD

- *ML data platform, ETL pipelines and CI/CD for fleet-scale models*
- Built and maintained production ETL pipelines on AWS (SageMaker, Lambda, Glue, S3) and Snowflake that fed continuous training, evaluation and deployment.
- Built Spark jobs orchestrated with Airflow for fleet itinerary management.
- Automated infrastructure provisioning and model deployment with Concourse CI/CD, keeping development, test and production environments reproducible.
- Combined SMB fleet customers' driving data with NZTA vehicle-inspection records to model vehicle status and driving behaviour.

Stack: Python · PyTorch · AWS (SageMaker, Lambda, Glue, S3) · Spark · Airflow · Snowflake · Concourse · Docker

Machine Learning Engineer

Jun 2021 — Feb 2022

StayinFront Ltd

- Developed backend services and ML models for detecting price tags and SKU information from shelf and packaging images.
- Forecast sales for supermarket groups such as Walmart from shelf images, inventory, pricing and shelf-life data, and integrated the results with StayinFront's ERP and the clients' ERP systems.
- Built the data pipeline behind the forecasts: PySpark processing and Python / Flask services connected to the existing AWS stack (S3, EC2, DynamoDB).

Stack: Python · Keras · TensorFlow · PySpark · DynamoDB · Flask · AWS

Machine Learning Engineer (Contract)

Sep 2020 — Apr 2021

BEDC (BuildingEstimates.com)

- Built a plan-recognition and building-information extraction system for architectural drawings (elevation, floor and roof plans), assessing building plans for quantity surveying.
- Recognised and processed plans with CNN and LSTM deep-learning models combined with PCA and HMM.
- Designed the framework architecture for building-material analysis and quantity-survey information.
- Developed the building-plan management platform in Python (Flask, pandas) and helped implement it on GCP, with Flask services processing plan images from Cloud Storage.

Stack: Python · pandas · NumPy · Flask · Git · GCP · Kubernetes

Data Engineer

Sep 2017 — Sep 2019

Woolworths NZ · Woolworths Group ASX:WOW

- Added random forest, CNN, RNN and MDAV algorithms to the data platform, with documented code.
- Integrated SAP and other ERP platforms through REST APIs for data-warehouse, logistics and payment workflows.
- Built cross-border payment and logistics solutions with a React Native web UI.
- Implemented and managed data procedures for security and integrity; Agile / Scrum delivery with CI/CD and testing specialists, versioned in GitHub.

Stack: Python · C# · TensorFlow · MS SQL Server · Docker · AWS · React Native · AngularJS

Full Stack Developer · New Image International — Jun 2014 — May 2017

E-commerce and ERP platforms: order processing, customer management, payment-gateway integration.

Software Developer · Huawei Symantec Technology — Dec 2010 — Aug 2013

Led frontend and backend teams; multithreaded Linux C/C++; time-series anomaly detection in spectrum data (Kalman filter, HMM).

Software Developer · Topsec Network Security (002212.SZ) — Oct 2007 — Mar 2010

Linux intrusion detection with kernel-level packet analysis; DDoS detection.

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